



Laxmi Charitable Trust's

Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce
Dr. S. Radhakrishnan Marg, Andheri (East) Mumbai 400 069



Department of Information Technology (Bsc.IT Semester III)
Python Programming Practical

Mini Project PDS

Roll No: 44	Name: Amit Umarvaish
Class: SYIT	Batch: 00

Project Report

1. Title of Project:- REAL ESTATE ANALYTICS DASHBOARD

Tools Used: Python, NumPy, Pandas, Matplotlib, Seaborn, Beautiful Soup, Tkinter

1. Introduction

Real estate data analysis helps us understand the property market by studying property prices, locations, property types, area, bedrooms and construction status. In this project, a real estate dataset containing 100 property records from five cities (Mumbai, Navi Mumbai, Thane, Pune and Nagpur) is analyzed using Python libraries. Different statistical methods and visualizations are used to identify market patterns, and the results are presented through an interactive Real Estate Analytics Dashboard.

2. Objective

- To analyze real estate data using Python.
- To load, clean and organize the dataset using Pandas.

- To perform numerical and statistical analysis using NumPy and Pandas.
- To create meaningful visualizations using Matplotlib and Seaborn.
- To collect sample web data using Beautiful Soup.
- To present important results using a Real Estate Analytics Dashboard built with Tkinter.

3. Dataset Description

The real estate dataset contains 100 records and 8 columns.

Column	Description
Property_ID	Unique ID of the property
Date	Date of the property record
Location	City where the property is located
Property_Type	Type of property (Apartment, Villa, Penthouse, Studio)
Bedrooms	Number of bedrooms
Area_sqft	Property area in square feet
Price_Lakh	Property price in Lakh (₹)
Status	Ready to Move / Under Construction

4. Tools and Libraries Used

Library	Purpose
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NumPy	Numerical calculations (sum, mean) and array operations
Pandas	Dataset loading, cleaning and analysis
Matplotlib	Data visualization (line chart, donut chart)
Seaborn	Statistical visualization (bar plots)
Beautiful Soup	Extracting data from HTML/web pages
Tkinter	Creating the dashboard window and layout

5. Data Cleaning

Cleaning operations performed:

1. Removed duplicate records using `drop_duplicates()`.
2. Converted the Date column into proper date format using `pd.to_datetime()`.
3. Handled missing values in numerical columns (Bedrooms, Area_sqft, Price_Lakh) using their mean values.
4. Removed extra spaces from column names using `str.strip()`.

Code:-

```
file = os.path.join(os.path.dirname(__file__), "real_estate_dataset.csv")
df = pd.read_csv(file)
df.columns = df.columns.str.strip()
df = df.drop_duplicates()
df["Date"] = pd.to_datetime(df["Date"])
for c in ["Bedrooms", "Area_sqft", "Price_Lakh"]:
    df[c] = df[c].fillna(df[c].mean())
```

```

print("Data Cleaning Completed!")
print("\nShape of Dataset:", df.shape)
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())
print("\nDuplicate Records After Cleaning:")
print(df.duplicated().sum())

```

Output:-

```

Data Cleaning Completed!
Shape of Dataset: (100, 8)
Missing Values After Cleaning:
Property ID      0
Date             0
Location         0
Property_Type    0
Bedrooms         0
Area_sqft        0
Price_Lakh       0
Status           0
dtype: int64
Duplicate Records After Cleaning:
0

```

6. Data Analysis

Code:-

```

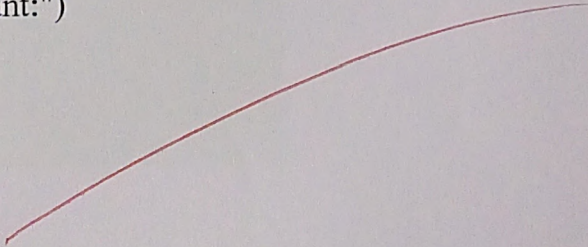
total = np.sum(df.Price_Lakh)
avg = np.mean(df.Price_Lakh)
top_loc = df.groupby("Location").Price_Lakh.sum().idxmax()
top_type = df.Property_Type.value_counts().idxmax()

loc =
df.groupby("Location").Price_Lakh.sum().sort_values(ascending=False)
month = df.groupby(df.Date.dt.month).Price_Lakh.sum()
ptype = df.Property_Type.value_counts()

```

```
bins = [0,100,200,300,500,np.inf]
labels = ["<100L","100-200L","200-300L","300-500L","500L+"]
price =
pd.cut(df.Price_Lakh,bins,labels=labels).value_counts().sort_index()
```

```
print("DATA ANALYSIS RESULTS")
print("-----")
print("Total Value:", f"₹{total:,.0f} L")
print("Average Price:", f"₹{avg:,.1f} L")
print("Total Listings:", len(df))
print("Top Location:", top_loc)
print("Top Property Type:", top_type)
print("\nProperty Value by Location (Lakh):")
print(loc)
print("\nMonthly Property Value (Lakh):")
print(month)
print("\nProperty Type Count:")
print(ptype)
print("\nPrice Range Count:")
print(price)
```



Output:-

```
DATA ANALYSIS RESULTS
.....
Total Value: ₹19,727 L
Average Price: ₹197.3 L
Total Listings: 100
Top Location: Navi Mumbai
Top Property Type: Villa

Property Value by Location (Lakh):
Location
Navi Mumbai    5805
Nagpur          4077
Mumbai          3981
Pune            3710
Thane           2154
Name: Price_Lakh, dtype: int64

Monthly Property Value (Lakh):
Date
1    6607
2    5687
3    5293
4    2140
Name: Price_Lakh, dtype: int64

Property Type Count:
Property Type
Villa          32
Apartment      23
Penthouse      23
Studio         22
Name: count, dtype: int64

Price Range Count:
Price Lakh
<100L          15
100-200L       38
200-300L       29
300-500L       18
500L+           0
Name: count, dtype: int64
```

Web Data Extraction using BeautifulSoup

Code:-

```
soup =
BeautifulSoup("<div>Mumbai</div><div>Pune</div><div>Thane</div>
>", "html.parser")
web_data = [x.text for x in soup.find_all("div")]
```

```
print("Web Data Extracted:", web_data)
```

Output:-

```
Web Data Extracted: ['Mumbai', 'Pune', 'Thane']
```

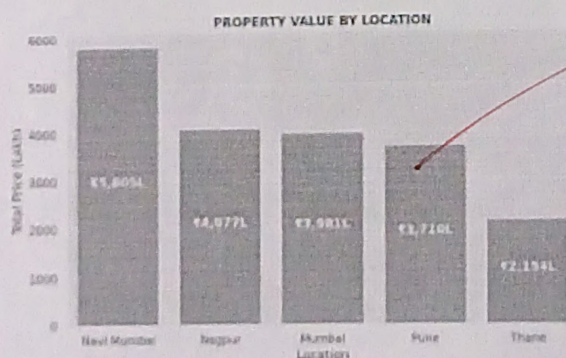
7. Data Visualizations

A. Bar Chart – Property Value by Location

Code:-

```
plt.figure(figsize=(8, 5))
sns.barplot(x=loc.index, y=loc.values, color="#4f8cff")
plt.title("PROPERTY VALUE BY LOCATION", fontweight="bold")
plt.xlabel("Location")
plt.ylabel("Total Price (Lakh)")
for i, v in enumerate(loc.values):
    plt.text(i, v/2, f'₹{v:,.0f}L', ha="center", color="white",
             fontweight="bold")
plt.tight_layout()
plt.show()
```

Output:-

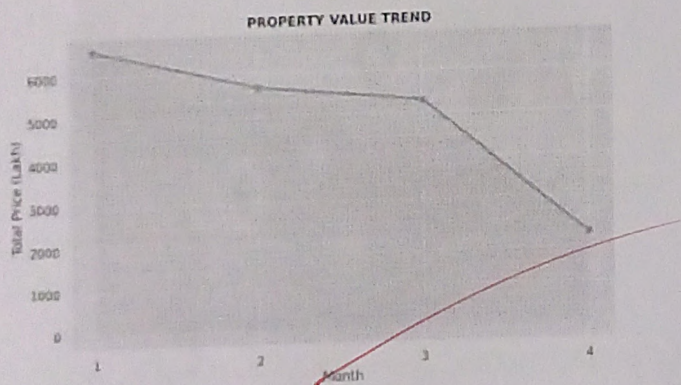


B. Line Chart – Property Value Trend

Code:-

```
plt.figure(figsize=(8, 5))
plt.plot(month.index, month.values, color="#22c7d9", marker="o",
linewidth=2.5)
plt.fill_between(month.index, month.values, color="#22c7d9",
alpha=.12)
plt.title("PROPERTY VALUE TREND", fontweight="bold")
plt.xlabel("Month")
plt.ylabel("Total Price (Lakh)")
plt.xticks(month.index)
plt.tight_layout()
plt.show()
```

Output:-



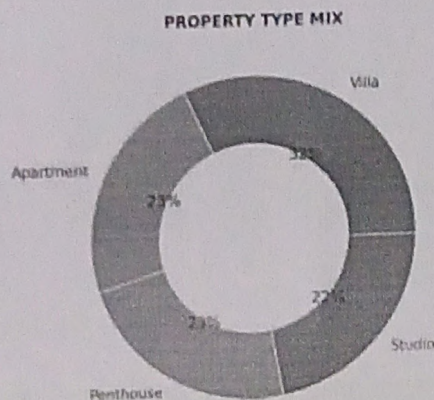
C. Donut Chart – Property Type

Mix Code:-

```
plt.figure(figsize=(6, 5))
plt.pie(pdtype.values, labels=pdtype.index, autopct="%1.0f%%",
```

```
wedgeprops={"width": .42, "edgecolor": "white"})
plt.title("PROPERTY TYPE MIX", fontweight="bold")
plt.tight_layout()
plt.show()
```

Output:-

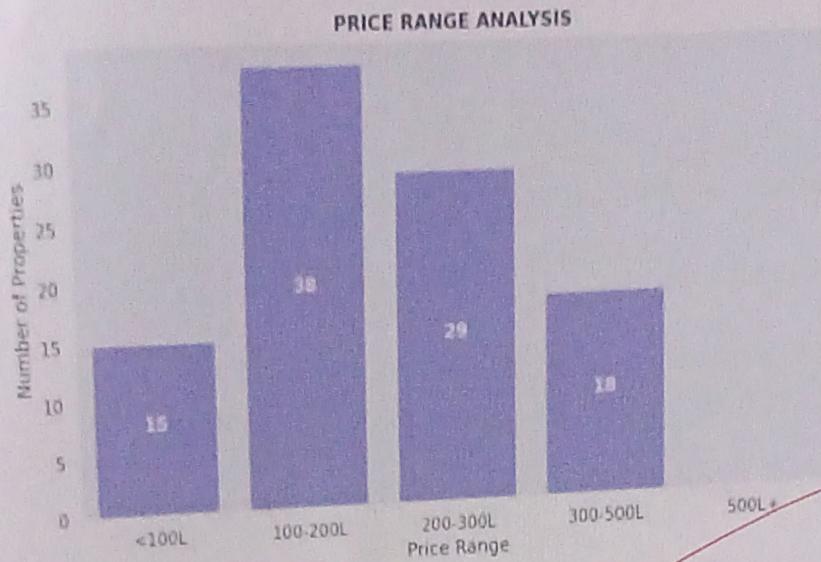


D. Bar Chart – Price Range Analysis

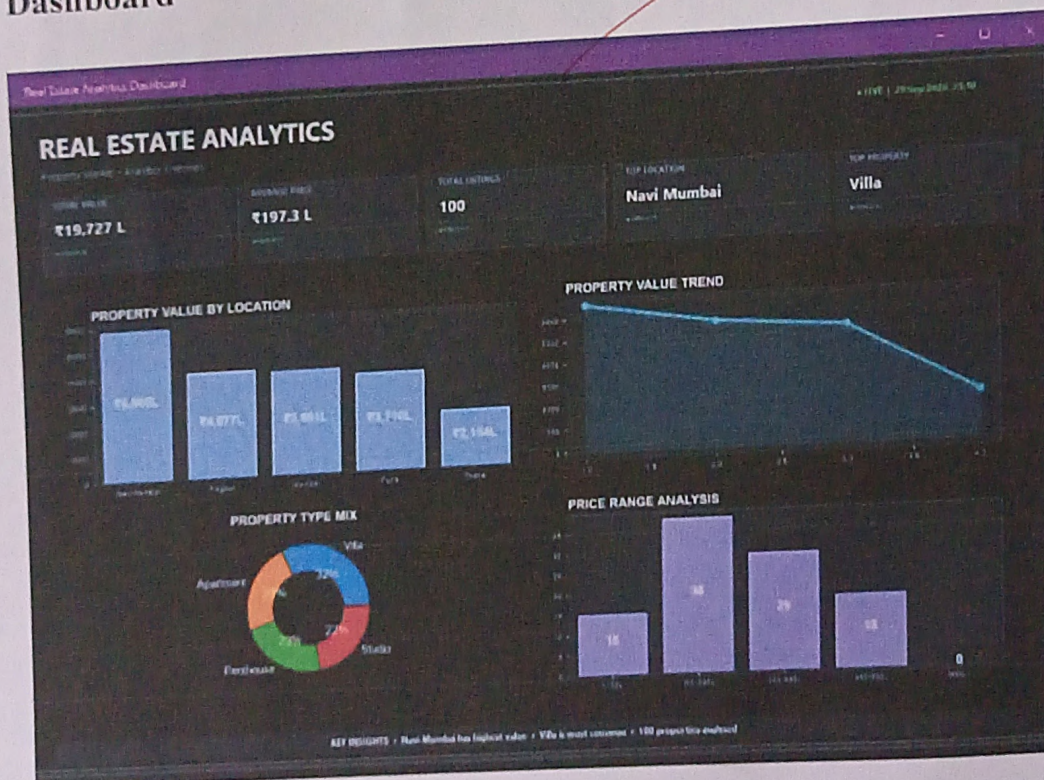
Code:-

```
plt.figure(figsize=(8, 5))
sns.barplot(x=price.index, y=price.values, color="#8b6cff")
plt.title("PRICE RANGE ANALYSIS", fontweight="bold")
plt.xlabel("Price Range")
plt.ylabel("Number of Properties")
for i, v in enumerate(price.values):
    plt.text(i, v/2, str(v), ha="center", color="white", fontweight="bold")
plt.tight_layout()
plt.show()
```

Output:-



8. Dashboard



9. Key Findings/Observations

The dataset contains 100 real estate records from 5 locations.

The dataset contains 4 different property types: Villa, Apartment, Penthouse and Studio.

The total value of all properties is ₹19,727 Lakh and the average price is approximately ₹197.3 Lakh.

The highest property price is ₹349 Lakh and the lowest is ₹35 Lakh.

Navi Mumbai has the highest total property value (₹5,805 Lakh), while Thane has the lowest (₹2,154 Lakh).

Villa is the most common property type (32%), followed by Apartment and Penthouse (23% each) and Studio (22%).

The 100–200 Lakh price range has the most properties (38), and no property is priced above 500 Lakh.

Out of 100 properties, 58 are Ready to Move and 42 are Under Construction.

The monthly trend shows the highest value in January (₹6,607 Lakh) and the lowest in April (₹2,140 Lakh); April contains records only up to 10 April.

All these results are presented together in the Tkinter dashboard using KPI cards and four charts.

10. Conclusion

The Real Estate Analytics Dashboard project successfully analyzed property market data using Python. Pandas and NumPy were used for data loading, cleaning and analysis, while Matplotlib and Seaborn were used to create meaningful visualizations such as bar charts, a line chart and a donut chart. Beautiful Soup was used for basic web data extraction from HTML content, and Tkinter was used to build the final dashboard window. The dashboard presents the important statistics, visualizations and key insights in an easy-to-understand format.

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